Data Exploration, Preparation and Mean-Prediction Baseline

Concrete is one of the most widely used construction materials in civil engineering. Its compressive strength is an important indicator of its ability to withstand applied loads without failure.

Concrete strength depends on several factors, including the quantities of cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate and fine aggregate as well as the concrete curing age.

The purpose of this project is to explore the concrete dataset prepare it for machine learning and establish a baseline model for predicting compressive strength using the mean-prediction approach.

Dataset Description

The supplied Concrete_Data.csv dataset was examined using Python and Pandas.

It contained the following ; Total obervations 1,030, Total variables were 9, independent variables were 8,Target variable was only 1, Missing values were 0, Duplicate values were 25and Numerical variables were 9 and categorical variables were 0.

Description of Dataset Variables

The dataset contains eight independent variables i.e Cement, Blast Furnace Slag, Fly Ash, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate, Age and one dependent variable which is Concrete Compressive Strength. The target variable is Concrete Compressive Strength, while the remaining eight variables are used as predictors.

Initial Data Exploration

The dataset was examined to understand its structure, variable types, distributions and general characteristics. All the nine variables are numerical. Eight variables are continuous measurements, while age is recorded as an integer representing the number of curing days.

The descriptive statistics indicated that the concrete samples cover a wide range of mixture compositions and curing periods. The compressive strength ranges from approximately 2.33 MPa to 82.60 MPa, indicating substantial variation in the strength of the concrete samples.

Distribution and Data Patterns

Some important patterns were observed during the exploratory analysis:

1. Cement: Cement quantities vary considerably, suggesting that different concrete mixtures use different cement proportions.

2. Blast Furnace Slag and Fly Ash: Some observations have zero quantities, indicating that these supplementary materials are not included in every mixture.

3. Water: Water quantities vary across observations and may influence concrete strength.

4. Superplasticizer: Several observations contain zero superplasticizer, while others contain varying quantities.

5. Age: Curing age ranges from 1 to 365 days. This is an important variable because concrete generally gains strength as curing progresses.

6. Compressive Strength: Strength measurements demonstrate considerable variability, suggesting that a simple linear or constant prediction approach may not adequately capture the underlying relationships.

Relationship Between Predictors and Concrete Strength. The correlation analysis provides an initial understanding of the linear relationships between individual predictors and compressive strength.

The findings suggest that: Cement has the strongest positive linear correlation with compressive strength, Superplasticizer and curing age also demonstrate positive relationships with strength, Water has a negative linear correlation with compressive strength and Fly ash, coarse aggregate and fine aggregate show relatively weak negative correlations.

NB: Important observation: Correlation measures linear association. It does not establish causation, and weak correlation does not necessarily mean that a variable is unimportant in predicting concrete strength. Nonlinear relationships and interactions between ingredients may be important.

Data Preparation

The dataset was checked for missing or null values and the examination revealed that all nine variables contain zero missing values and therefore no imputation was necessary because the dataset was complete. Applying unnecessary imputation could introduce artificial values and distort the original data distribution.

The analysis identified 25 exact duplicate rows. Duplicate records can lead to repeated representation of particular observations and may cause data leakage when identical records appear in both training and testing datasets and these were removed before splitting the dataset and after removing the 25 duplicates, the dataset contains 1,005 observations.

Handling Outliers

The Interquartile Range (IQR) method was considered for identifying potential outliers.

Some observations were identified as potential outliers, particularly in age, water and superplasticizer quantities. Preparation decision: The identified observations should not automatically be deleted. Extreme values may represent legitimate concrete mixtures or curing experiments. They should be investigated and retained unless there is evidence of measurement or recording errors.

Feature Scaling

The independent variables have different measurement ranges. For example, cement ranges from 102 to 540 kg/m³, whereas superplasticizer ranges from 0 to 32.2 kg/m³ hence feature scaling maybe necessary. However, for the mean-prediction baseline, scaling is unnecessary because the model does not use individual predictor values.

NB: When developing subsequent predictive models, the scaler should be fitted using training data only to prevent data leakage.

2.5 Training and Testing Data

The cleaned dataset was divided into training and testing sets using an 80:20 ratio.

A random state of 42 was used to ensure reproducibility.

The training dataset is used to establish the baseline prediction, while the testing dataset evaluates its predictive performance.

Establishing a Mean-Prediction Baseline. A mean-prediction baseline is a simple predictive model that estimates the target variable using the average value of the target variable in the training dataset.

In this project, the model predicts concrete compressive strength using the mean compressive strength of the training observations. The baseline does not consider the individual concrete ingredients or curing age.

Its purpose is to establish a minimum reference performance against which more advanced machine learning models can be compared.

Mean-Prediction Baseline Results. After removing duplicate records and splitting the dataset into training and testing sets, the mean-prediction baseline produced the following results.

Baseline evaluation completed

Evaluation measure	Result
Training observations	804
Testing observations	201
Training mean	35.0668 MPa
MAE	13.9723 MPa
MSE	298.9696 MPa²
RMSE	17.2965 MPa
R-squared	-0.0028

4.1 Interpretation of Evaluation Metrics

The baseline MAE is approximately 13.97 MPa and it means that predictions differ from actual concrete strength measurements by an average absolute amount of approximately 13.97 MPa.

Mean Squared Error (MSE). The MSE is approximately 298.97 MPa^2. It penalises larger prediction errors more heavily than smaller ones.

Root Mean Squared Error (RMSE). The RMSE is approximately 17.30 MPa. This indicates considerable prediction error when using a constant mean prediction.

Coefficient of Determination (R^2). The R^2 value is approximately -0.0028.

A slightly negative R^2 indicates that the model performs marginally worse than predicting the average strength of the testing observations. This is expected because the baseline does not consider the relationships between concrete ingredients, curing age and compressive strength.

Findings

The exploratory analysis demonstrates that concrete compressive strength depends on a combination of mixture composition and curing age.
The baseline results demonstrate that the average compressive strength alone cannot adequately explain the differences in concrete strength across the observations.

Conclusion and Recommendations

These results establish a reference point for evaluating more advanced predictive approaches. i.e Develop a multiple linear regression model to establish a more informative benchmark, Apply Random Forest Regression to capture nonlinear relationships, Evaluate Gradient Boosting or XGBoost to improve prediction accuracy, Investigate the influence of curing age and cement quantity on compressive strength, Compare models using MAE, RMSE and R^2 and Use cross-validation to evaluate model stability and reduce dependence on a single train-test split.